

a read-only multi-set of data items distributed over a cluster of machines, which is maintained in a fault-tolerant way. It was developed in response to limitations in the MapReduce cluster computing paradigm, which forces a particular linear data flow structure on distributed programs. MapReduce programs read input data from the disk, map a function across the data, reduce the results of the map, and store reduction results on the disk. Spark's RDDs function as a working set for distributed programs that offer a (deliberately) restricted form of distributed shared memory.

Databases

PostgreSQL is essentially a relational database, but with an object-oriented database model. Performance is nearly as fast as MariaDB and MySQL, but the biggest plus is PostgreSQL's management interface called pgAdmin, which has a robust set of features that's easy and intuitive to navigate.

MariaDB is a community-developed fork of the MySQL relational database management system intended to remain free under the GNU GPL. It is notable for being led by the original developers of MySQL, who forked it due to concerns over its acquisition by Oracle. Contributors are required to share their copyright with the MariaDB Foundation. It intends to maintain high compatibility with MySQL, ensuring a 'drop-in' replacement compatibility with library binary equivalence and exactly matches with MySQL APIs and commands. It includes the XtraDB storage engine for replacing InnoDB, as well as a new storage engine, Aria, that is intended to be both a transactional and non-transactional engine, perhaps even to be included in future versions of MySQL.

MongoDB is the NoSQL database movement that came about to address the shortcomings of relational databases and the demands of modern software development. MongoDB is the leading NoSQL database, with significant adoption among the Fortune 500 and Global 500 companies.

Application runtime environments

OpenJDK, PHP runtime and Apache Tomcat are the most widely used open source application runtime environments based on the kind of workloads enterprise class IT runs.

OpenJDK (Open Java Development Kit) is a free and open source implementation of the Java platform, Standard Edition (Java SE). The implementation is licensed under the GNU General Public License (GNU GPL) version 2 with a linking exception. OpenJDK is the official reference implementation of Java SE since version 7.

The OpenJDK project produces a number of components, most importantly, the virtual machine (HotSpot), the Java Class Library and the Java compiler (javac).

The Web browser plugin and Web Start, which form part of Oracle Java, are not included in OpenJDK. Sun previously indicated that it would try to open source these components, but neither Sun nor Oracle have done so. The only currently available free plugin and Web Start implementations, as of 2016, are those provided by IcedTea.

Apache Tomcat, often referred to as Tomcat Server, is an open source Java Servlet Container developed by the Apache Software Foundation (ASF). Tomcat implements several Java EE specifications including Java Servlet, JavaServer Pages (JSP), Java EL and WebSocket, providing a 'pure Java' HTTP Web server environment in which Java code can run.

Tomcat is developed and maintained by an open community of developers under the auspices of the Apache Software Foundation, released under the Apache License 2.0, and is open source software.

PHP is a server-side scripting language designed primarily for Web development but is also used as a general-purpose programming language. PHP originally stood for Personal Home Page, but it now stands for the recursive acronym PHP: Hypertext Preprocessor.

PHP code may be embedded into HTML code, or it can be used in combination with various Web template systems, Web content management systems and Web frameworks. PHP code is usually processed by a PHP interpreter implemented as a module in the Web server or as a Common Gateway Interface (CGI) executable. The Web server combines the results of the interpreted and executed PHP code, which may be any type of data, including images, with the generated Web page. PHP code may also be executed with a command-line interface (CLI) and can be used to implement standalone graphical applications.

The standard PHP interpreter, powered by the Zend Engine, is free software released under the PHP licence. PHP has been widely ported, and can be deployed on most Web servers on almost every operating system and platform, free of charge.

Manageability software

OpenDCIM is a free, Web based data centre infrastructure management (DCIM) application. While proprietary products dominate the DCIM solutions, OpenDCIM is the most promising open source alternative.

While OpenDCIM doesn't provide all the features available in commercial products, it covers a majority of them. OpenDCIM is not designed to compete with commercial products. Its biggest advantage is that it is effectively free for use. For data centres of almost any size, the tool is an effective introduction to DCIM concepts, and allows administrators to evaluate the utility of DCIM software in their own environments.

Open source hardware in the enterprise

Open compute project (OCP)

The Open Compute Project (OCP) is an organisation that shares designs of data centre products among companies, including Facebook, Intel, Google, Apple, Microsoft, Seagate Technology, Dell, Rackspace, Ericsson, Cisco, Juniper Networks, Goldman Sachs, Fidelity, Lenovo and Bank of America. It is a collaborative community focused on redesigning hardware technology to efficiently support the